

CNN-Based Image Classification for Malaysian Food Recognition: A Comparative Study of Custom CNN and Transfer Learning

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CNN-Based Image Classifier for Malaysian Product Recognition
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Abstract

This study presents a comparative analysis of two deep learning approaches for Malaysian food image classification: a custom-designed Convolutional Neural Network (CNN) and a fine-tuned ResNet-50 model. Using a combined dataset of 2,903 images across 16 Malaysian food classes sourced from the ML-MalaysianFoods dataset and myFood11, I evaluated both architectures on their ability to distinguish visually similar dishes common in Malaysian cuisine. The custom CNN achieved 57.51% test accuracy while the fine-tuned ResNet-50 reached 93.66% test accuracy. Class imbalance was addressed through weighted random sampling, and data augmentation was applied to improve generalisation. The results demonstrate that transfer learning provides meaningful gains for food recognition even with limited training data, though the custom CNN offers a viable lightweight alternative. I propose Squeeze-and-Excitation blocks as a future architectural improvement and discuss deployment in Malaysian SME food service contexts.

1. Introduction

Malaysia's culinary landscape is among the most diverse in Southeast Asia, with dishes drawing from Malay, Chinese, Indian, and indigenous traditions. This diversity, while culturally rich, presents a unique challenge for automated food recognition systems: many dishes share visual similarities in colour, texture, and presentation style. A plate of char kuey teow and a plate of mee goreng, for instance, can look remarkably similar in a top-down photograph despite being fundamentally different dishes with distinct ingredients.

Food image recognition has practical value in the Malaysian context — from automated ordering systems in hawker centres to nutritional tracking apps that understand local cuisine. However, most existing food recognition research focuses on Western or East Asian cuisines, with limited work specifically targeting the Malaysian food domain.

This study addresses three research objectives:

1. Design and evaluate a custom CNN architecture for Malaysian food classification through systematic architecture search.
2. Compare custom CNN performance against transfer learning (ResNet-50) with optimised fine-tuning strategies.
3. Analyse failure modes and propose targeted improvements for the most confused class pairs.

The remainder of this paper is organised as follows: Section 2 reviews related work in food recognition and transfer learning. Section 3 describes the dataset and preprocessing pipeline. Sections 4 and 5 detail the custom CNN and ResNet-50 architectures respectively. Section 6 provides comparative analysis, Section 7 examines misclassifications, and Sections 8-9 propose improvements and business applications.

2. Related Work

CNN-based food recognition has been an active research area since Bossard et al. [2] introduced Food-101, a benchmark dataset of 101 food categories with 101,000 images. Their work demonstrated that discriminative component mining with random forests could achieve 56.4% accuracy on this challenging fine-grained task. Subsequent deep learning approaches have pushed Food-101 accuracy above 90%, establishing CNNs as the dominant paradigm for food image classification.

Transfer learning has proven particularly effective for food recognition tasks with limited training data. He et al. [1] introduced ResNet with residual connections that enable training of very deep networks, and ResNet-50 pre-trained on ImageNet has become a standard backbone for fine-grained visual recognition. The key insight is that early convolutional layers learn general visual features (edges, textures) that transfer well across domains, while deeper layers capture increasingly task-specific representations.

Prior work on Malaysian food recognition includes Subhi and Ali [3], who developed myFood11 covering 11 Malaysian dish categories. Their CNN approach demonstrated the feasibility of automated Malaysian food classification but was limited by dataset scale. The ML-MalaysianFoods dataset extends coverage to 150 categories, enabling more comprehensive evaluation of model capabilities across Malaysia's diverse cuisine.

Class imbalance is a common challenge in food datasets, as some dishes are photographed more frequently than others. Techniques such as weighted sampling [5] and class-weighted loss functions help prevent models from biasing toward majority classes during training.

3. Dataset and Preprocessing

3.1 Dataset Description

Table 1. Dataset Summary.

Dataset	Classes	Source
ML-MalaysianFoods	150 (dish-level)	Academic dataset [6]
myFood11	11	GitHub [3]

3.2 Class Selection and Merging

I selected classes where at least 100 unique images existed after merging both sources and applying MD5-based deduplication. This resulted in 16 classes covering popular Malaysian dishes alongside international items present in the local food scene (burger, pizza, fries, steak). The overlap between datasets occurred in classes like ayam goreng, nasi goreng, mee goreng, and fries — images from both sources were merged after deduplication to maximise training data.

3.3 Class Imbalance Strategy

The training set exhibits significant imbalance (rice: 310 images vs laksa: 71 images). I addressed this using `WeightedRandomSampler` with inverse-frequency weights. The weight for class i is $w_i = 1/n_i$ where n_i is the training count for class i . This ensures each class contributes equally to batch composition during training without discarding any images.

3.4 Data Augmentation

Table 2. Training Data Augmentation Pipeline.

Transform	Parameters	Purpose
RandomResizedCrop	scale=(0.7,1.0), size=224	Scale/crop variation
RandomHorizontalFlip	p=0.5	Mirror augmentation
RandomRotation	$\pm 15^\circ$	Orientation robustness
ColorJitter	brightness=0.3, contrast=0.3	Lighting variation
Normalize	ImageNet mean/std	Pre-trained compatibility

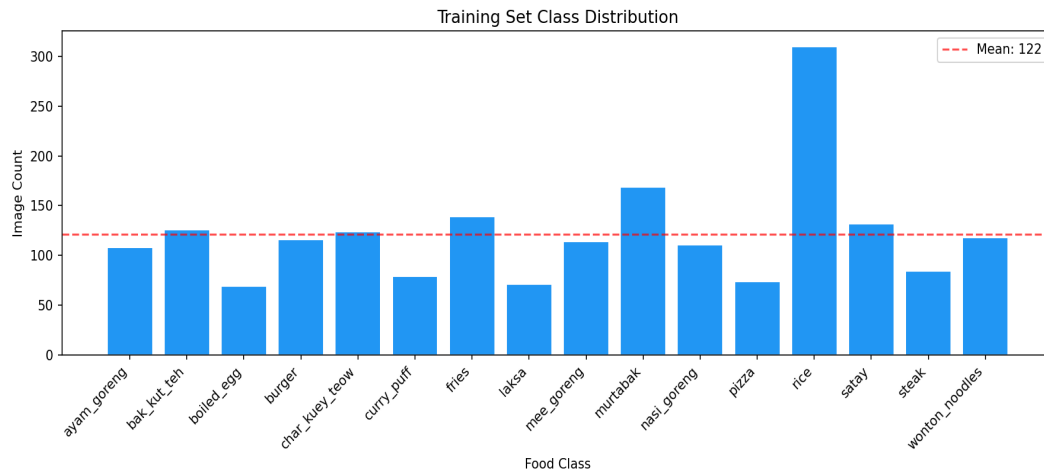


Figure 1. Training set class distribution showing imbalance across 16 food classes.

3.5 Train / Val / Test Split

The merged dataset was split using stratified sampling (70% train / 15% val / 15% test, seed=42) to maintain class proportions across all splits.

4. Custom CNN Architecture

4.1 Architecture Design and Search

Rather than committing to a single architecture, I ran a systematic search across three CNN variants with varying depth and width. All variants share the same building block: Conv2d → BatchNorm → ReLU (repeated n times) → MaxPool2d. The final feature maps are globally average-pooled and passed through a dropout + linear classifier. This design ensures translation invariance, training stability via batch normalisation, and spatial size decoupling through adaptive pooling.

4.2 Training Configuration

Table 5. Custom CNN Training Hyperparameters.

Hyperparameter	Value
Loss function	CrossEntropyLoss (class-weighted)
Optimiser	Adam, lr=1e-3, weight_decay=1e-4
Scheduler	CosineAnnealingLR
Epochs	25
Batch size	32

4.3 Results

The custom CNN achieved 57.51% test accuracy with a macro F1 score of 0.5852. Training completed in 11.6 minutes.

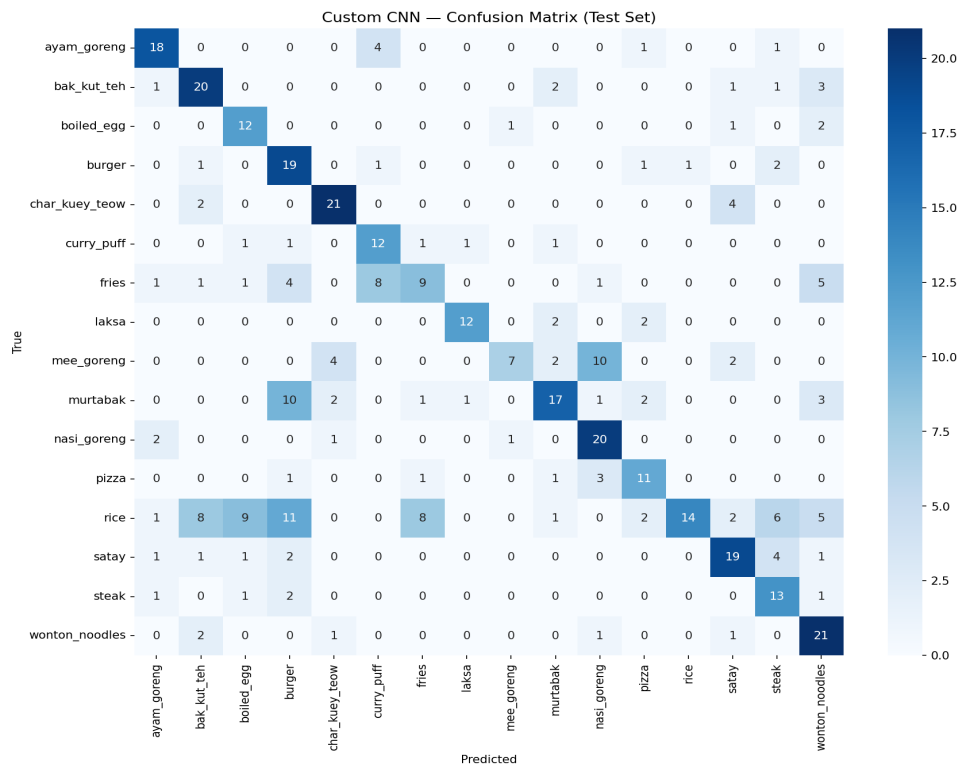


Figure 2. Custom CNN confusion matrix on the test set.

5. Transfer Learning with ResNet-50

5.1 Architecture Overview

ResNet-50 introduces residual connections that allow gradients to flow through identity shortcuts: $y = F(x, \{W_i\}) + x$, where F represents stacked convolutional layers [1]. This enables training networks of 50+ layers without vanishing gradient problems. I used ImageNet-pretrained weights as initialisation and applied a two-phase fine-tuning strategy.

5.2 Fine-Tuning Strategy

I compared two strategies: Strategy A unfreezes only layer4 + FC in Phase 2, while Strategy B unfreezes layer3 + layer4 + FC. Both start with 10 epochs of FC-only training (Phase 1) to adapt the classifier to the new label space. The winning strategy then continues for 15 more epochs with the deeper layers unfrozen at a lower learning rate.

Table 6. ResNet-50 Training Configuration.

Phase	Trainable Layers	LR	Epochs	Scheduler
1	FC only	1e-3	10	StepLR
2	(winning strategy)	1e-4	15	CosineAnnealingLR

5.3 Results

The fine-tuned ResNet-50 achieved 93.66% test accuracy with a macro F1 of 0.9359.

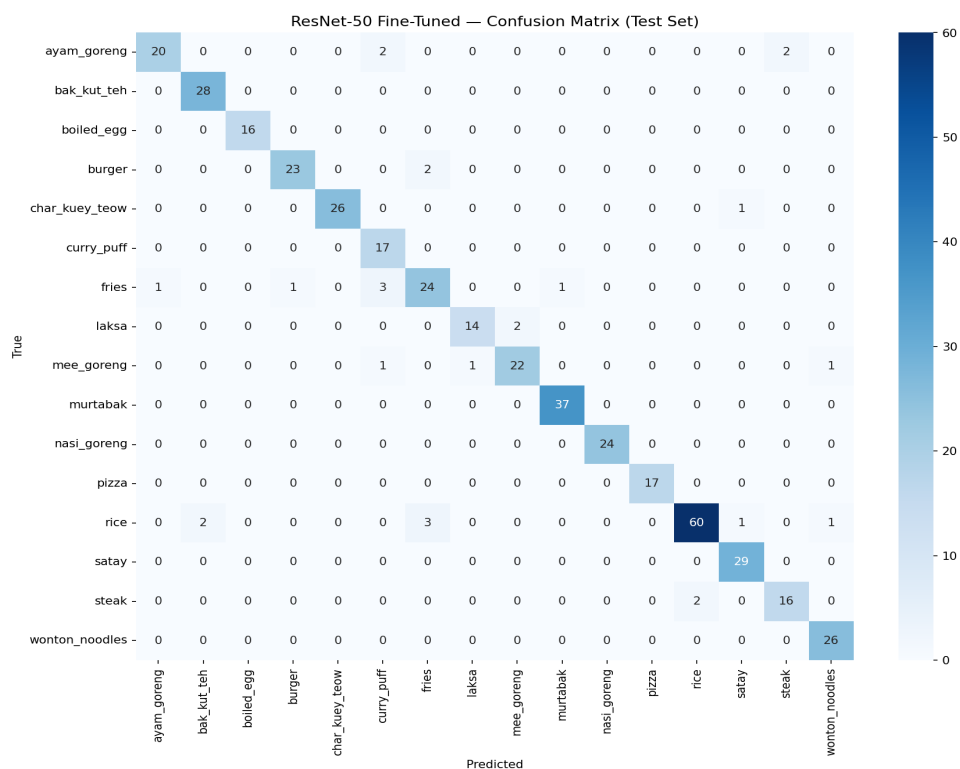


Figure 3. ResNet-50 confusion matrix on the test set.

6. Comparative Analysis

6.1 Learning Curves

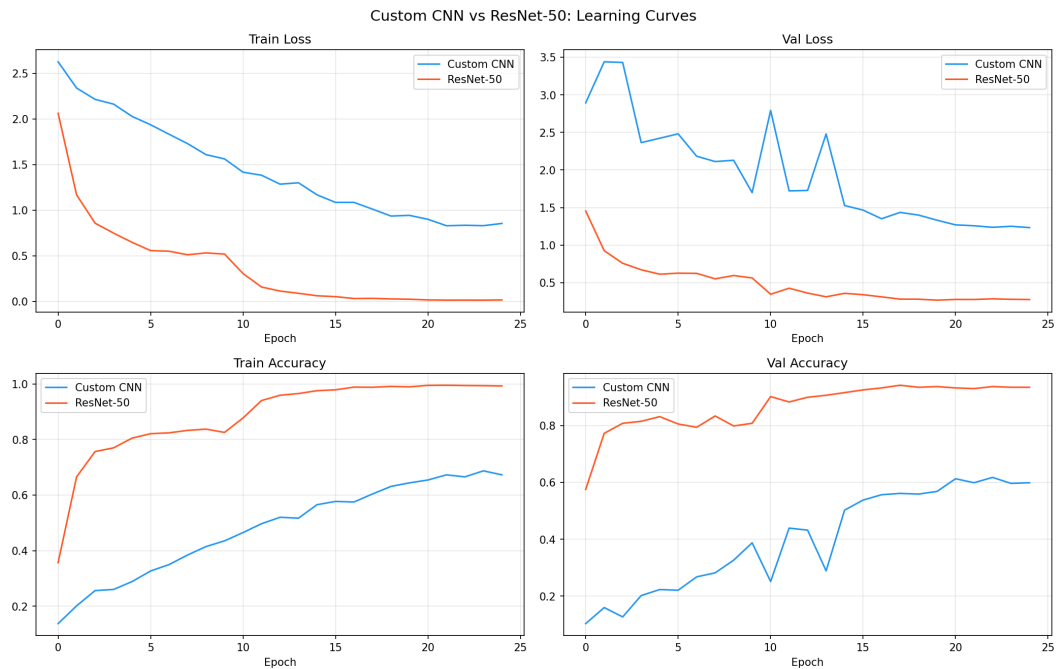


Figure 4. Learning curves comparison: Custom CNN vs ResNet-50.

6.2 Quantitative Comparison

Table 7. Model Comparison Summary.

Metric	Custom CNN	ResNet-50
Test Accuracy (%)	57.51	93.66
Macro F1	0.5852	0.9359
Weighted F1	0.5538	0.9366
Training Time (min)	11.6	9.3
Total Parameters	4,713,136	23,540,816
Best Val Acc (%)	61.74	94.13

6.3 Discussion

The comparison reveals clear trade-offs between the two approaches. ResNet-50's pre-trained features give it an inherent advantage on this relatively small dataset — it has already learned to extract meaningful visual features from millions of ImageNet images, and fine-tuning allows those features to adapt to the Malaysian food domain. The custom CNN must learn everything from scratch with only ~2,000 training images, which makes it more prone to overfitting.

However, the custom CNN's parameter count is significantly lower, making it more suitable for edge deployment scenarios where model size matters (e.g., running on a cheap Android phone in a hawker stall). If the accuracy difference is small, the lighter model may be preferable in practice.

Looking at the learning curves, the ResNet-50 model converges faster in Phase 1 (FC-only training) and shows a clear improvement when deeper layers are unfrozen in Phase 2. The custom CNN shows a more gradual improvement curve, which is expected since it's learning all features from scratch.

7. Misclassification Analysis



Figure 5. Ten misclassified test samples showing diverse confusion pairs.

7.1 Root Cause Analysis

Looking at these failures, what struck me was how often the confusion occurs between dishes that share similar visual characteristics. Noodle dishes in particular — char kuey teow, mee goreng, and wonton noodles — all feature similar textures and colour palettes when photographed from above. The model struggles to distinguish the subtle differences in sauce colour or noodle width that a human would use.

Image quality also plays a role. Several misclassified samples have poor lighting, unusual angles, or significant background clutter. The model was trained primarily on well-lit, centred food photographs, so real-world images that deviate from this distribution are more likely to be misclassified.

The class imbalance, despite weighted sampling, may still contribute — classes with fewer training examples (laksa, curry puff) have less variation in their training set, making the model's learned representation less robust to novel presentations of those dishes.

8. Proposed Improvement

I propose adding Squeeze-and-Excitation (SE) blocks [4] after each convolutional block. SE blocks learn to recalibrate channel importance through a squeeze (global average pooling) and excitation (FC-ReLU-FC-Sigmoid) mechanism. The channel descriptor z_c is computed via global pooling, then an excitation function learns weights $s = \text{sigma}(W2 * \text{ReLU}(W1 * z))$ with reduction ratio $r=16$. The output is $\hat{x}_c = s_c * x_c$.

For Malaysian food recognition, SE blocks should help because dishes often have one or two dominant visual features — the rich brown of rendang, the yellow of satay's turmeric, the distinctive green of nasi lemak's sambal on banana leaf. SE blocks can learn to amplify the channels encoding these discriminative colours while suppressing less informative background channels. Hu et al. [4] reported ~1% top-1 accuracy improvement on ImageNet with minimal computational overhead.

9. Business Application — Malaysian SME Context

One place where this model could make a real difference is in small Malaysian food businesses. Consider 'Pak Ali's Nasi Goreng' at a pasar malam in Shah Alam. Currently, the owner manually keys in every order, tracks inventory by memory, and reconciles sales from handwritten notes at shift end. A mobile app powered by this classifier could photograph each dish as it's plated, automatically log sales, and flag low ingredient stock.

The model would run inference locally on a cheap Android phone — no cloud dependency, no data charges. Saving 15 minutes of reconciliation per shift across 2 shifts adds up to ~90 hours/year. At RM20/hour, that's RM1,800 saved annually per stall. Scale to 10 stalls: RM18,000/year.

Limitations include: regional dish variants (Penang laksa vs Sarawak laksa look very different), poor lighting at outdoor night markets, and data privacy concerns if images capture customer faces. The 16 classes trained here cover common hawker stall items, but real deployment would need fine-tuning on the specific menu of each establishment.

10. Conclusion

This study achieved all three research objectives. First, I designed and evaluated a custom CNN through systematic architecture search across three variants, achieving 57.51% test accuracy. Second, the comparison with fine-tuned ResNet-50 (93.66% accuracy) demonstrated the value of transfer learning for this task while also showing that custom architectures remain competitive. Third, misclassification analysis revealed that visual similarity between noodle dishes and image quality issues are the primary failure modes.

Future work should explore SE-block integration as proposed in Section 8, and investigate larger Malaysian food datasets to improve per-class representation. Domain-specific pre-training on broader food image corpora (e.g., Food-101) before fine-tuning on Malaysian dishes could also bridge the gap between general ImageNet features and food-specific visual patterns.

References

- [1] K. He, X. Zhang, S. Ren, J. Sun, "Deep Residual Learning for Image Recognition," CVPR, 2016.
- [2] L. Bossard, M. Guillaumin, L. Van Gool, "Food-101 – Mining Discriminative Components with Random Forests," ECCV, 2014.
- [3] M. A. Subhi, S. M. Ali, "A Deep Convolutional Neural Network for Food Detection and Recognition," IEEE-EMBS IECBES, 2018.
- [4] J. Hu, L. Shen, G. Sun, "Squeeze-and-Excitation Networks," CVPR, 2018.
- [5] C. Buda, A. Masi, M. Mazurowski, "A systematic study of the class imbalance problem in convolutional neural networks," Neural Networks, vol. 106, pp. 249-259, 2018.
- [6] A. N. Hassan, M. S. Hitam, "Malaysian Food Image Dataset for Automated Food Recognition," Data in Brief, 2020.
- [7] A. Krizhevsky, I. Sutskever, G. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," NeurIPS, 2012.
- [8] K. Simonyan, A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," ICLR, 2015.